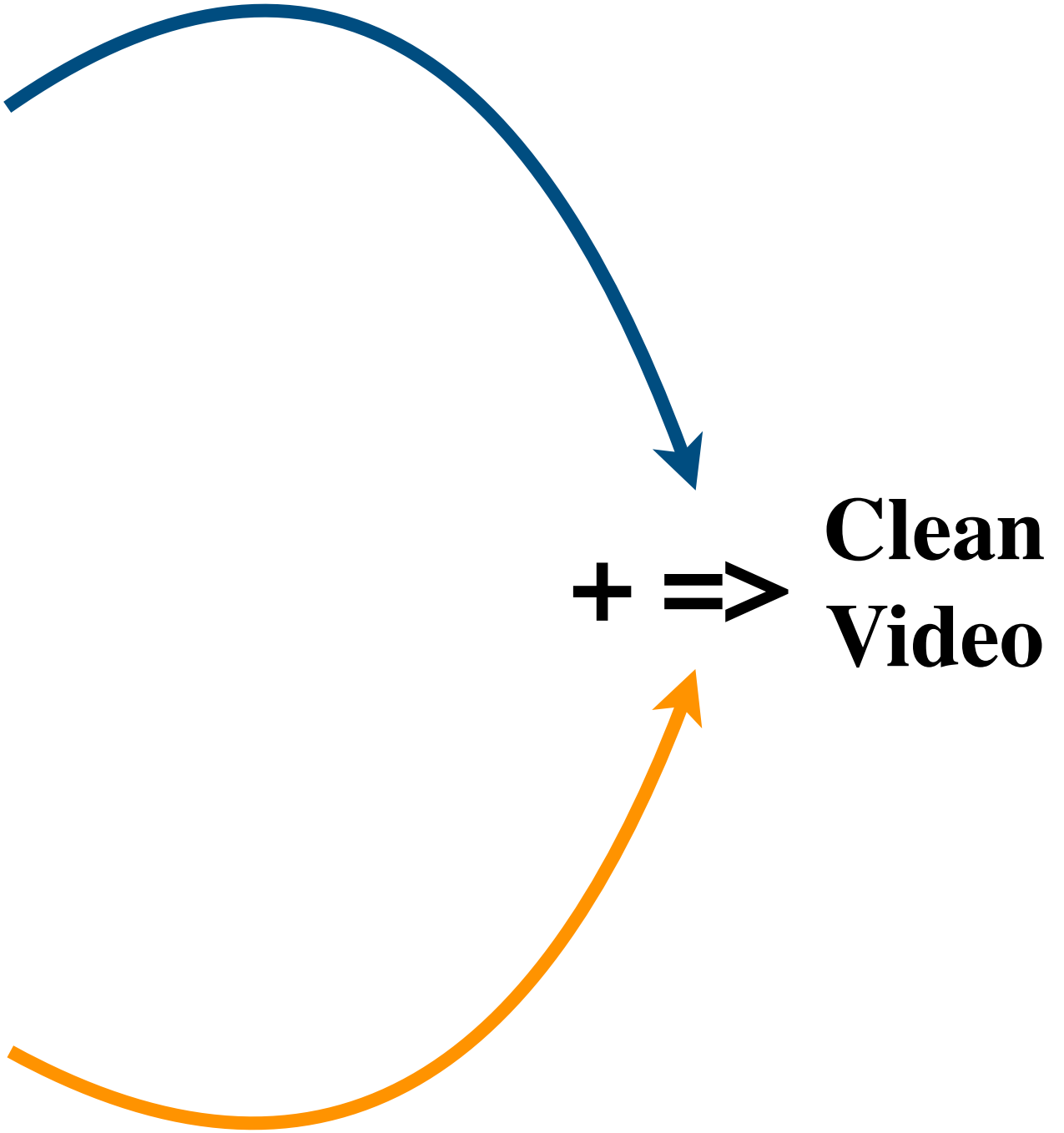
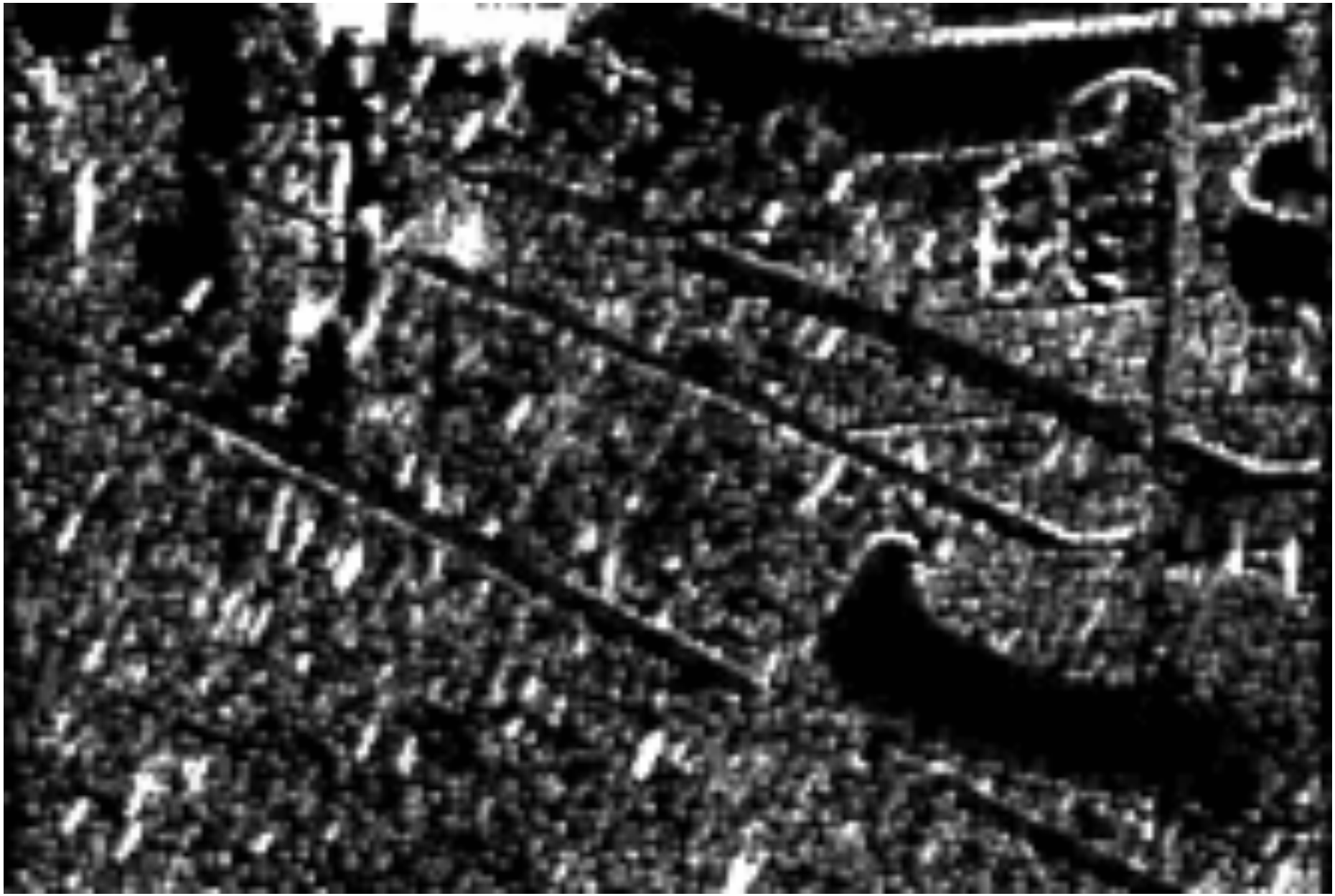

+ =>
Clean
Video

Background

Rain layer

Moving objects (Foreground)

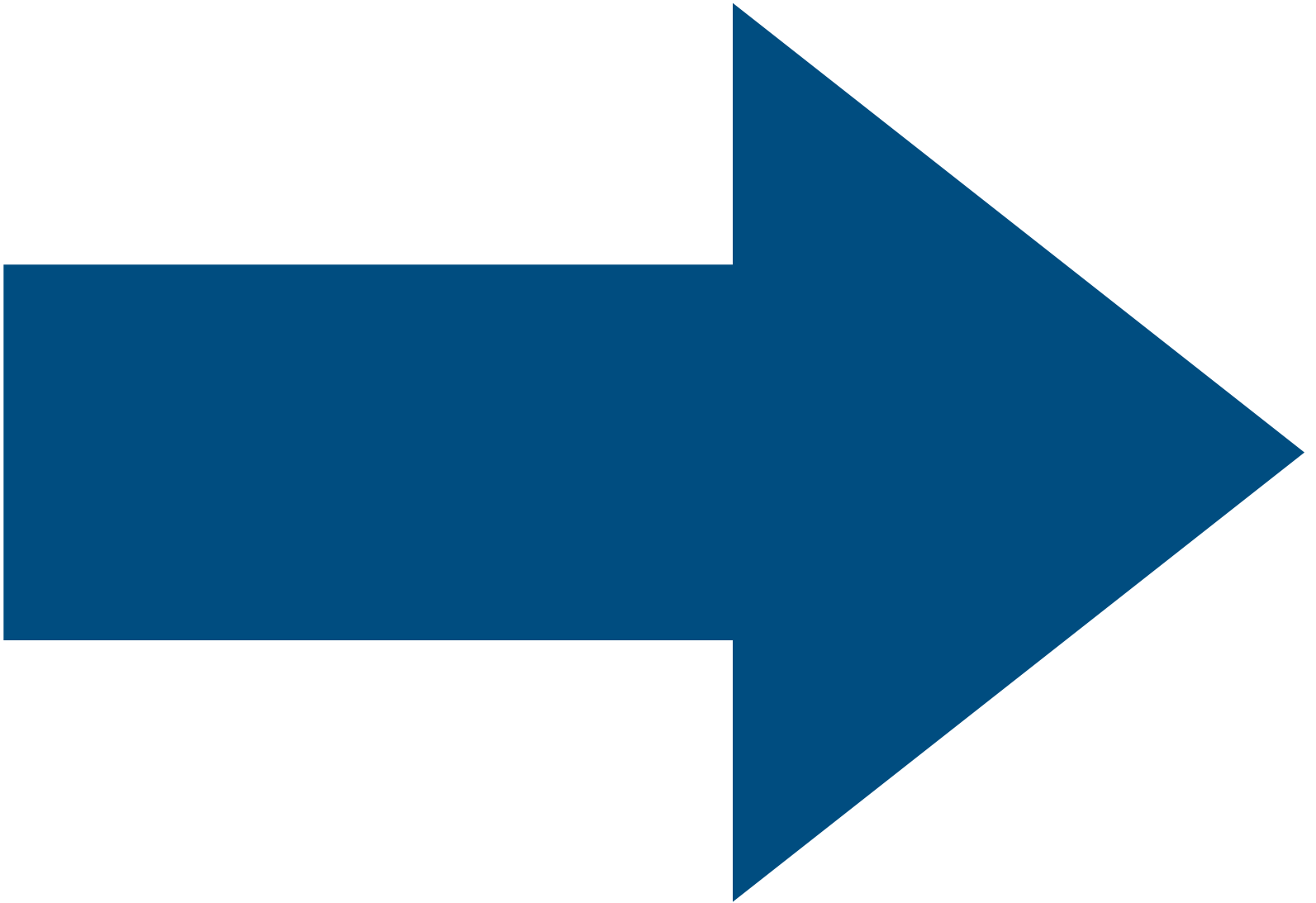


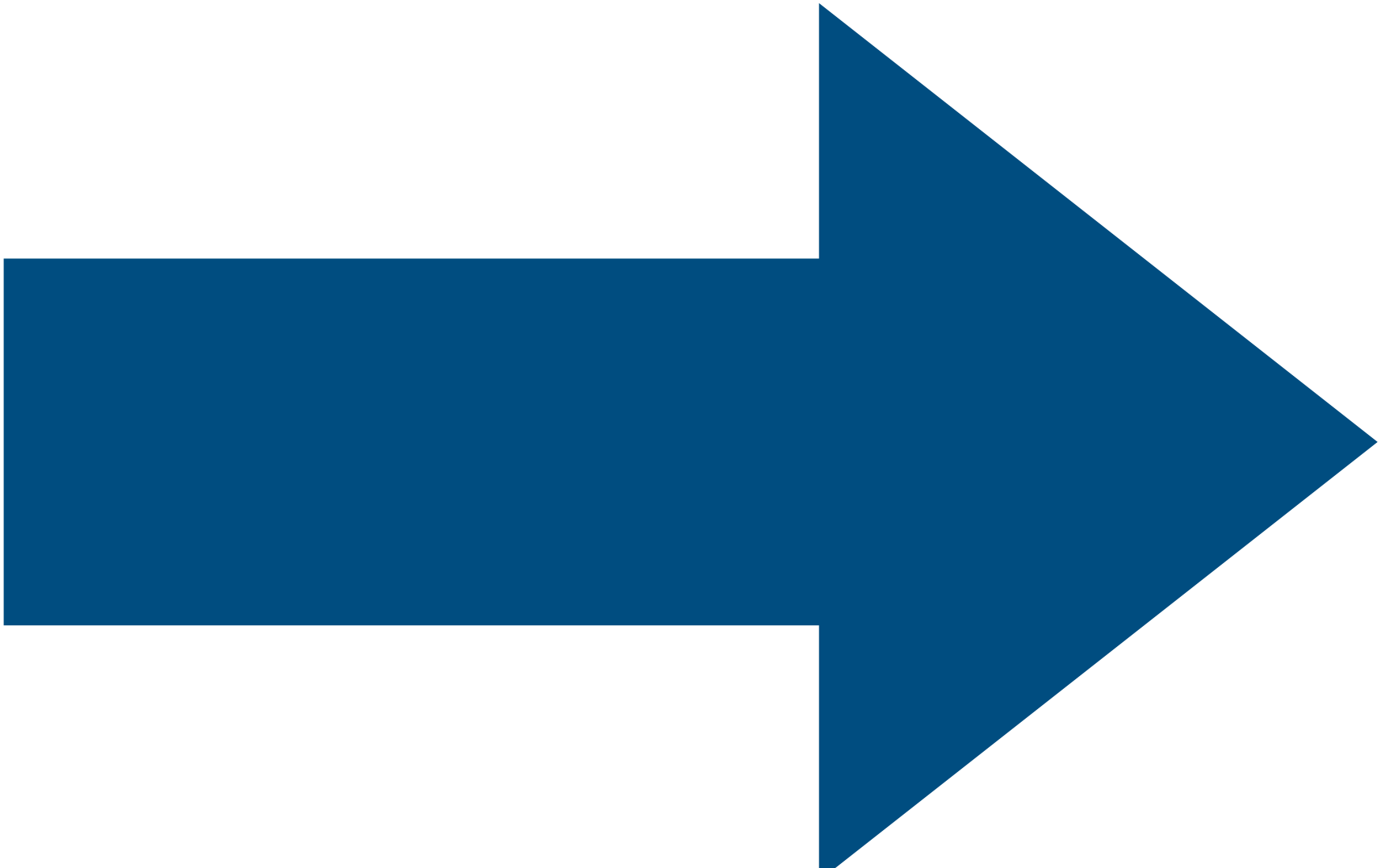
Rainy Video

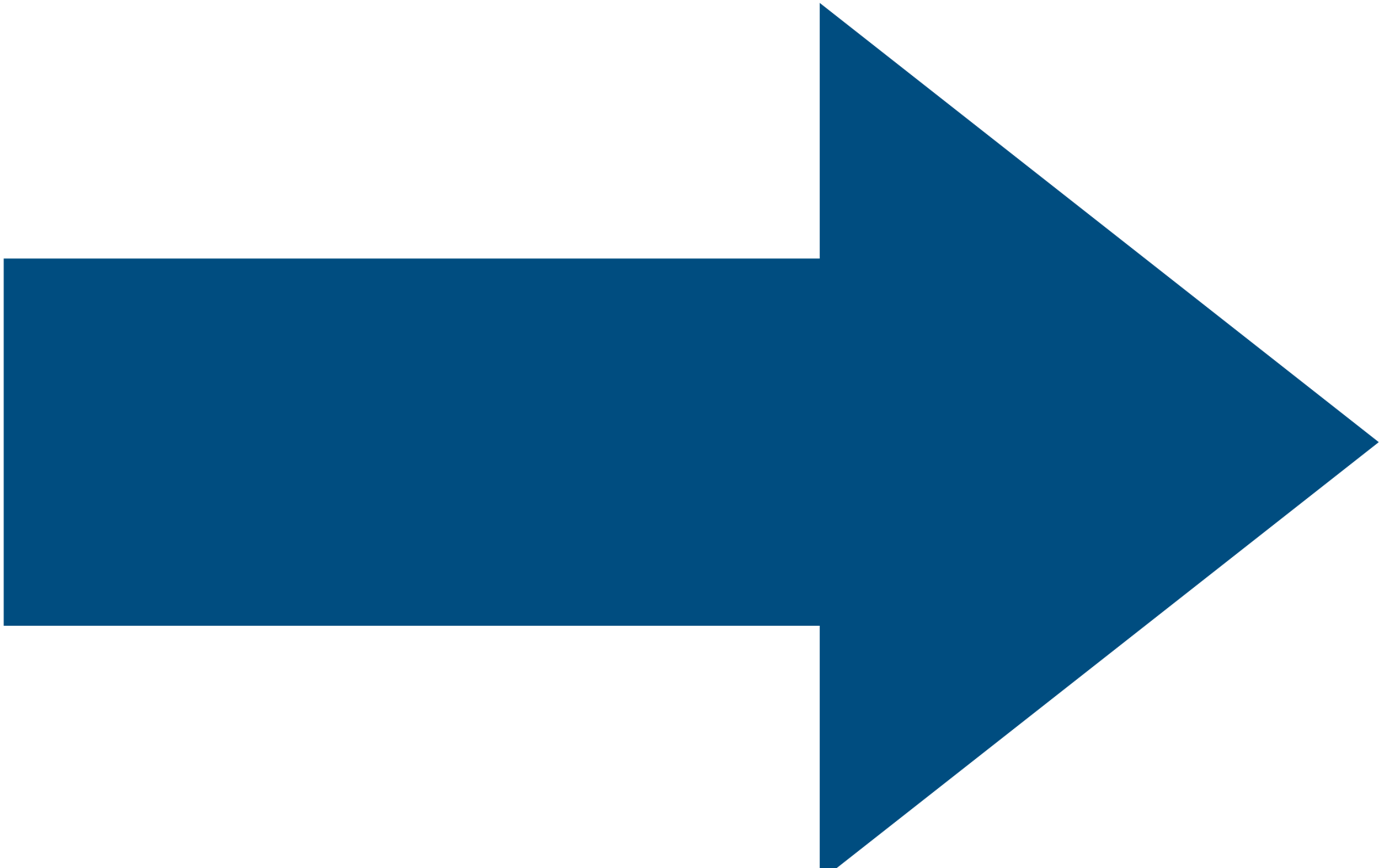
OT-MSCSC: Online Rain/Snow Removal from Surveillance Videos (TIP 2021). **First author, high-impact, used in real-world environments**

Video Rain/Snow Removal

1



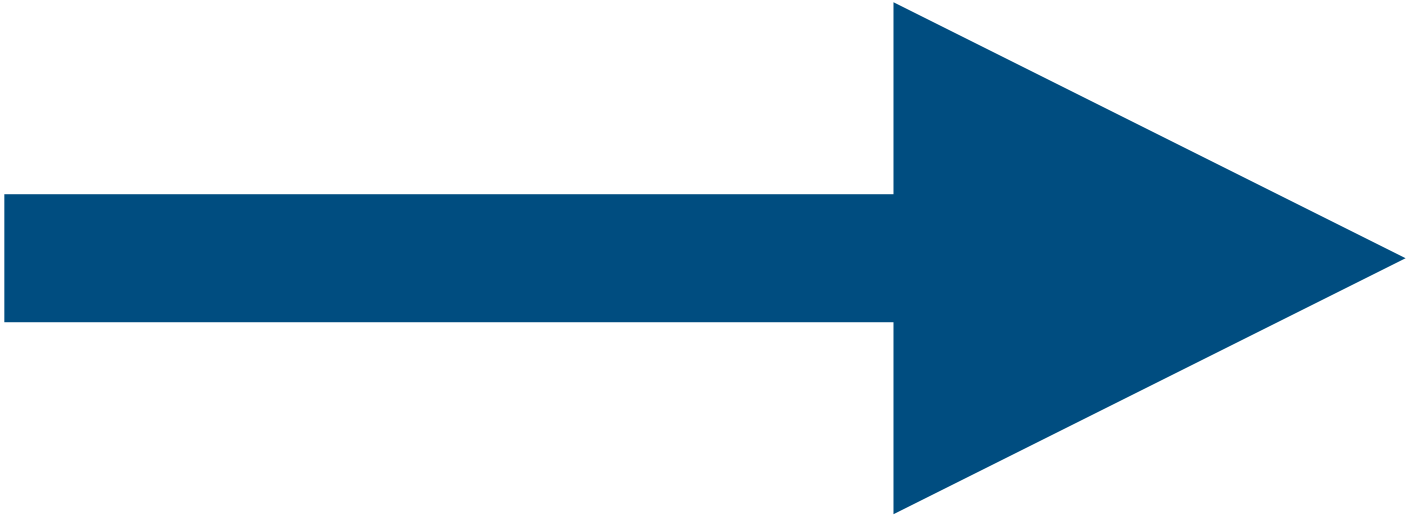




Noisy input

Clean output

Model



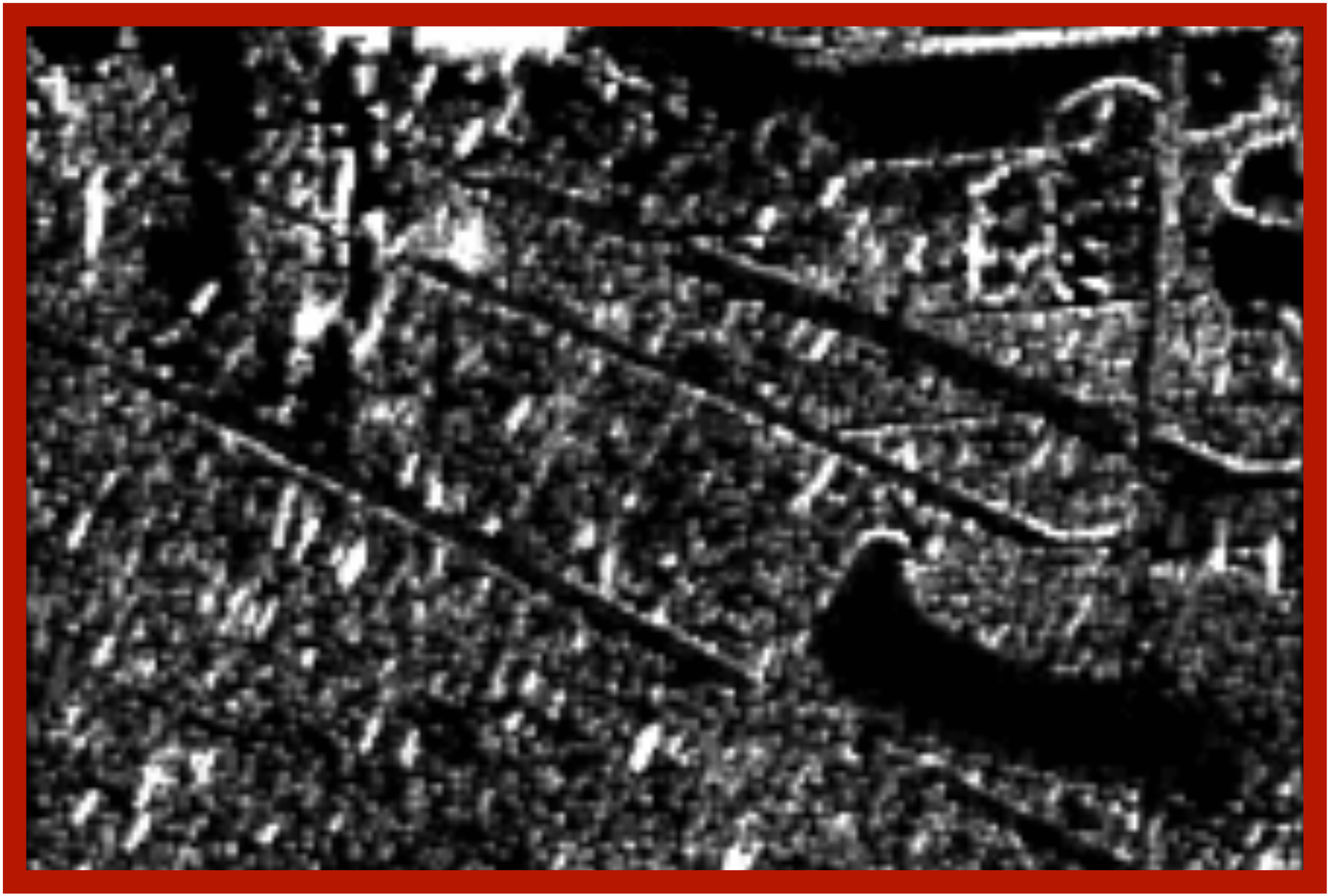
Key idea

Goal 1: completely remove rain or snow

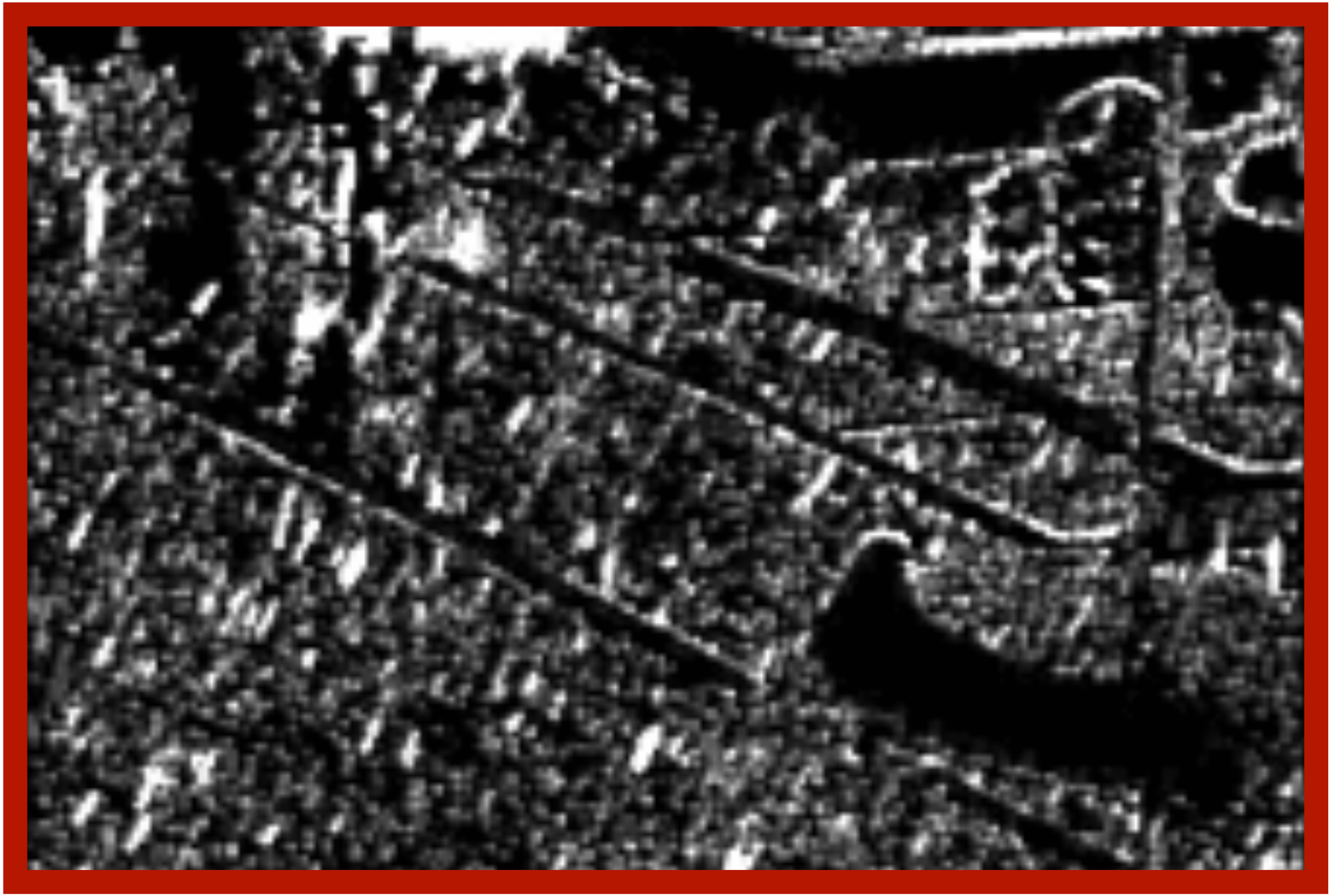
Goal 2: keep visual details unchanged

UNITED STATES
POSTAL SERVICE

UNITED STATES
POSTAL SERVICE



Rain layer



Moving objects

(Foreground)

Video

Video Decomposition

$$\mathcal{B} = \text{Fold}(UV^T),$$

$$U \in R^{d \times r}, V \in R^{n \times r},\ r < \min(d, n).$$

Low Rank Matrix Decomposition

redundancy

Temporal

$$\mathcal{R} = \sum_{k=1}^{K} \sum_{s=1}^{n_k} D_{ks} \otimes \mathcal{M}_{ks}.$$

Sparse &

repetitive patterns

at multiple scales

Multi-scale Convolutional Sparse Coding

Markov Random Filed:

$$\mathcal{H}_{ijn} = \begin{cases} 1, & \text{location } ijn \text{ is moving objects,} \\ 0, & \text{location } ijn \text{ is background.} \end{cases}$$

$$\mathcal{H} \circ \mathcal{X} = \mathcal{H} \circ (\mathcal{F} + \mathcal{R}).$$

consistency over

time

Motion

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Model

Goal 1: completely remove rain or snow

Goal 2: keep visual details unchanged

Each component follows certain physical or statistical properties!

Background

Moving objects

(Foreground)